

Curriculum Vitae

PERSONAL INFORMATION

Walter Lucas Monteiro de Mendonça, Ph.D.

Senior Software Engineer | Senior Java Developer | AWS Certified
Brasília, DF, Brazil

Tel.: +55 (61) 99538-8569

Email: waltimlmm@gmail.com

LinkedIn: <https://www.linkedin.com/in/walter-lucas-mendonca/>

Homepage: <https://waltim.github.io/>

GitHub: <https://github.com/waltim>

ORCID: <https://orcid.org/0000-0001-7391-9622>

PROFESSIONAL SUMMARY

I am a Software Engineer and Senior Java Developer with over 10 years of experience building, maintaining, and modernizing backend applications. I have a strong background in the Java ecosystem (from versions 11 to 21) and Spring Boot, designing and evolving microservices architectures for critical systems. I bring to the industry the solid analytical foundation of my Ph.D. in Computer Science, applying source code rejuvenation and static analysis to optimize endpoints, eliminate technical debt, and ensure high test coverage in CI/CD pipelines. I hold the AWS Certified AI Practitioner certification and continuously seek to integrate cloud and AI innovations into scalable products.

PROFESSIONAL AND ACADEMIC EXPERIENCES

2024 - Current

Fóton Informática S.A., Brasília, Brazil

Position: Senior Software Engineer / Java Developer

Project: Designed and implemented scalable microservices for a major bank's real estate financing application using Java 11/17 and Spring Framework. Integrated services via IBM ACE and API Gateway, with deployment in OpenShift environments. Optimized critical endpoints, achieving up to 10x reduction in response time through targeted refactoring. Applied strict static analysis (SonarLint, Fortify) and implemented automated testing (JUnit, Mockito) ensuring high code coverage in CI/CD pipelines.

Technologies: Java 11/17, Spring Boot, Microservices, OpenShift, CI/CD, JUnit, Mockito, SonarLint, Fortify, Oracle, REST APIs, SCRUM.

2023 - 2024

University of Brasília, UnB, Brasília, Brazil

Position: Backend Developer and Product Owner (Virtual Social Office - Phase 2)

Project: Led the development team and defined the software architecture of the Virtual Social Office (ESV). Built robust APIs and backend services using Java 21 and Spring Boot 3. Coordinated the mobile app migration to React Native (Expo) and implemented a modern authorization system. Ensured LGPD compliance and conducted Google Analytics feasibility studies.

Technologies: Java 21, Spring Boot 3, Microservices, Node.js, React Native, Docker, LGPD Compliance, REST API, Scrum.

2022 - 2023

INESC TEC International Visiting Researcher Programme, Portugal

Position: Researcher

Project: Conducted an in-depth study analyzing GitHub repository histories of 272 projects to understand KDE contributor practices when adopting modern C++ features.

Research topic: Software engineering, C++ evolution, Software rejuvenation, Mining Software Repositories.

2020 - 2022	University of Brasília / Finatec, Brasília, Brazil Position: Data Engineer and Developer (Virtual Social Office - Phase 1) Project: Built and maintained REST APIs and microservices architecture using Java 17 to support the ESVirtual application data flow. Developed ETL pipelines using Python to consolidate critical data. Led the technical expansion of the pilot project to multiple states, automating business rule validations in spreadsheets to eliminate manual bottlenecks. Technologies: Java 17, Microservices, REST API, Python, ETL, Data Management, Information Security.
2018 - 2020	Department of Electrical Engineering, University of Brasília, Brazil Position: Data Engineer / Software Developer Project: Maintained and updated management dashboards using ETL tools like Pentaho and QlikView. Created a complete web system in PHP for strategic objective management. Configured and managed Rancher containers and built Jenkins CI/CD pipelines for automated deployments. Technologies: Java, PHP, Jenkins, Rancher, Pentaho, Qlik Sense, OLAP, CI/CD, ETL, Cloud Computing.
2017 - 2018	Finatec - PromiseEB Project, Brazil Position: Software Developer (Applied Research) Project: Acted in the reengineering of legacy systems for the Brazilian Army. Applied architecture visualization (CodeCity) and static analysis (iPlasma) to map technical debt. Developed in Java guided by Behavior-Driven Development (BDD) within a strict Scrum workflow. Technologies: Java, BDD, Static Analysis, Scrum, Reverse Engineering.
2015 - 2017	Mídia Proativa & MAXPHONE, Patos de Minas, MG, Brazil Position: Web Developer and Systems Analyst Project: Developed and managed e-commerce platforms, web systems, and point-of-sale systems using PHP (Laravel and CakePHP) and MySQL. Elicited requirements directly with clients, performed continuous website maintenance, and optimized product sales and web traffic using Google Analytics. Technologies: PHP, Laravel 5/6, CakePHP 2.x, MySQL, JavaScript, Google Analytics, Requirements Elicitation.

EDUCATION

2019 - 2026	Ph.D., Computer Science University of Brasília, UnB, Brazil Thesis: A Socio-Technical Grounded Theory about Source Code Rejuvenation
2017 - 2019	Master.Sc., Computer Science University of Brasília, UnB, Brazil Thesis: Analysis of the impact on the understanding of Java programs with the adoption of lambda expressions Awards: Best paper award at SBES 2019.
2011 - 2015	Bachelor's Degree, Information System University Center of Patos de Minas, UNIPAM, Brazil Awards: 7th Entrepreneurship Award (2015) and Unideias (2016).

PUBLICATIONS

Journal Publications

1. MENDONÇA, W. L. M., FORTES, J., LOPES, F. V., MARCÍLIO, D., DE ALMEIDA, R. B., CANEDO, E. D., LIMA, F., AND SARAIVA, J. Understanding the impact of introducing lambda

- expressions in java programs. *Journal of Software Engineering Research and Development* 8 (2020), 7–1
2. LUCAS, W., CARVALHO, F., NUNES, R. C., BONIFÁCIO, R., SARAIVA, J., AND ACCIOLY, P. Embracing modern c++ features: An empirical assessment on the kde community. *Journal of Software: Evolution and Process* 36, 5 (2024), e2605
 3. LUCAS, W., NUNES, R. C., BONIFÁCIO, R., CARVALHO, F., LIMA, R., SILVA, M., TORRES, A., ACCIOLY, P. R. G., MONTEIRO, E., AND SARAIVA, J. Understanding the adoption of modern javascript features: An empirical study on open-source systems. *Empir. Softw. Eng.* 30, 3 (2025), 107

Conferences

1. LUCAS, W., BONIFÁCIO, R., CANEDO, E. D., MARCÍLIO, D., AND LIMA, F. Does the introduction of lambda expressions improve the comprehension of java programs? In *Proceedings of the XXXIII Brazilian Symposium on Software Engineering* (2019), pp. 187–196
2. DE MENDONÇA, W. L. M., COSTA, P. H. T., CANÇADO, E. C. R., LIMA, F., CANEDO, E. D., BONIFÁCIO, R., AND AMARAL, L. H. V. From dusk till dawn: Reflections on the impact of covid-19 on the development practices of a r&d project. In *Proceedings of the 34th Brazilian Symposium on Software Engineering* (2020), pp. 596–605
3. DA FROTA, F. V. L., DE MENDONÇA, W. L. M., DA SILVA, D. A., DE SOUZA FERNANDES, M. A., BARBOSA, N. F., AND CANEDO, E. D. Utilização de algoritmos genéticos em gerações de dados de teste de software com agrupamento de genes e redução de espaço de busca. *WWW/INTERNET 2019*, 87

Extended Abstract / Short Papers

1. DANTAS, R., CARVALHO, A., MARCÍLIO, D., FANTIN, L., SILVA, U., LUCAS, W., AND BONIFÁCIO, R. Reconciling the past and the present: An empirical study on the application of source code transformations to automatically rejuvenate java programs. In *2018 IEEE 25th international conference on software analysis, evolution and reengineering (SANER)* (2018), IEEE, pp. 497–501
2. LUCAS, W., BONIFÁCIO, R., AND SARAIVA, J. Understanding the motivations, challenges, and practices of software rejuvenation. In *2023 IEEE International Conference on Software Maintenance and Evolution (ICSME)* (2023), pp. 611–616

LICENSES & CERTIFICATIONS

2026 AWS Certified AI Practitioner - Amazon Web Services (AWS)

LANGUAGES

Portuguese (Native)

English (Advanced, TOEFL ITP: 443)

Spanish (Intermediate)